MASTER THESIS · UPDATE 1

# Grading handwritten exams with vision-language models

Like ETH’s STACK, which checks typed math by meaning, but for handwritten answers

[Your name] · Supervisor meeting · 28 October 2026

WHAT EXISTS

# STACK checks meaning, but only for typed answers

TYPED + COMPUTER ALGEBRA

### STACK, Numbas, WeBWorK, PrairieLearn

Grades by mathematical meaning (Maxima): *2(x+1) = 2x+2*. Randomised questions, instant feedback.

Limit: students must type math.

HANDWRITTEN + AI ASSIST

### Gradescope (Turnitin)

AI groups similar handwritten answers; the instructor grades each group once.

Limit: a human still decides; fixed-template PDFs only.

RESEARCH 2025–26

### Vision-language model graders

Levine et al. 2026: 95% rubric-item accuracy on university STEM work; most errors come from **misreading** handwriting.

Limit: no meaning check like STACK.

Sources: ethz.ch STACK page · Gradescope docs · arXiv 2605.19043, 2606.11477

DATASETS BY DIFFICULTY LEVEL

# Public data exists for every level; math has the most

| Level | Dataset | Contents | Use |
|---|---|---|---|
| Fill the gaps | Grabowski 2026 | 61 exams, 3,141 handwritten answer boxes | Evaluate |
| Short answers | CREG (German) | 1,032 handwritten learner answers, 177 prompts | Evaluate |
| Free text | VisEssay (2022) | 13k+ graded handwritten essays (no public download) | Evaluate |
| Math | MathWriting, HME100K | 253k handwritten formulas → LaTeX; 99k photos | Fine-tune |
| Math | FERMAT (ACL 2025) | 2,244 handwritten solutions with labelled errors | Evaluate |
| Math | CHECK-MAT (2025) | 122 exam solutions with expert grades | Evaluate |

Not found: a public German fill-the-gaps worksheet set. A 3,945-record calculus set (2026) is not released.

PROPOSED APPROACH

# Fine-tune the reading step, then check like STACK

**1 · Capture**

Scan or photo of the answer sheet

**2 · Read**

Vision-language model → text / LaTeX. Fine-tune here.

**3 · Check**

SymPy equivalence, as in STACK; LLM for rubric text

**4 · Grade**

Score and feedback per step, teacher reviews

### Why step 2

For one model, 87% of grading errors came from misreading handwriting (Levine et al. 2026).

### Open models to compare

Qwen2.5-VL, olmOCR 2, PaddleOCR-VL, DeepSeek-OCR, Chandra OCR 2

### Reference points

Gemini, Mistral OCR (closed APIs); VEHME (fine-tuned grader, EMNLP 2025)

## DRAFT RESEARCH QUESTION

# Does fine-tuning an open vision-language model on handwritten math improve grading accuracy when its output is checked by computer algebra, STACK-style?

**Hypothesis:** fine-tuned reading + algebra check beats a zero-shot model grading directly, measured on FERMAT.

## NEXT TWO WEEKS

1. Download FERMAT and MathWriting; check licences
2. Zero-shot baseline: Qwen2.5-VL vs Gemini on a FERMAT subset
3. Prototype the SymPy equivalence check
4. Read 5 core papers closely, skim about 20

## QUESTIONS

- **Tasks:** math + fill-the-gaps with synthetic handwriting – OK?
- **Essays:** drop, or ask the VisEssay authors for data?
- **Seminar paper:** VEHME or Henkel et al.?